Band-1 Happy Ending / Erdős–Szekeres: ES(7) Lit/Box Increment

Finite re-certificates + cup-cap lowers + literature ledger refresh (NOT ES(7)=33)

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parent. `2026-09-11_Band1_Computational_Experiment_happy_ending` (lab `/workspace/experiments/band1/`
Lab. `/workspace/mission-control/experiments/band1/happy_ending_es7_increment/`
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite computational / literature ledger increment** only. It **does not** claim the Happy Ending / Erdős–Szekeres conjecture solved. It **does not** claim ES(7) = 33 proved. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Classical upper-bound literature is cited, not re-derived.

Abstract

We ship a Band-1 increment on the planar Erdős–Szekeres / Happy Ending numbers: (i) re-certify box exacts ES(3) = 3 and ES(4) = 5; (ii) rebuild classical double cup-cap lower witnesses of size 2^{n-2} with no convex n-gon for $n = 3, \ldots, 7$ (including an honest general-position size-32 witness for $n = 7$); (iii) refresh the literature ledger for ES(7) (exacts $3, 5, 9, 17$; conj. 33; arXiv:2512.24061 still **partial anchored UNSAT only** — verified against public sources on 2026-09-14). Optional checks: size-8 / size-16 witnesses have max_convex < 5 and < 6. The exact value ES(7) and the full conjecture remain **OPEN**.

Keywords. Erdős–Szekeres; Happy Ending; ES(7); cup-cap construction; finite box certificates; Band-1; non-claim.

1. Statement lock

Definition. ES(n) is the smallest integer M such that every set of M points in the plane in general position (no three collinear) contains n points in convex position.

Conjecture (Erdős–Szekeres / Happy Ending). $\mathrm{ES}(n) = 2^{n-2} + 1$ for all $n \geq 3$.

Known exacts (literature). ES(3) = 3, ES(4) = 5, ES(5) = 9, ES(6) = 17. **Next open exact:** ES(7) (conjectured 33).

Finite lemma (this increment only). Re-certified box exacts for $n = 3, 4$; verified classical lower witnesses for $n = 3..7$ of size 2^{n-2} with no convex n-gon (hence $\mathrm{ES}(n) \geq 2^{n-2} + 1$). **One finite ledger increment — not Happy Ending solved; not** ES(7) = 33**.**

2. What changed vs 2026-09-11

Axis	Prior (2026-09-11)	This increment
Lab path	`/workspace/experiments/band1/happy_ending/`	`/workspace/kissing-number-dimension-control/experiments/`
ES(3), ES(4)	certified	**re-certified** (smoke + grid/tri2)
Lowers $n = 3..7$	built	**rebuilt + timed**; $n = 7$ size 32 genpos, maxC = 6
Lit ledger	2026-09-11	**refreshed 2026-09-14** with web check
arXiv:2512.24061	cited as partial anchored UNSAT	**re-verified still partial** (no full ES(7)=33)
Claim level	finite / lit	**same; ES(7) OPEN**

3. Unit tests

Geometry primitives reused from parent `es_exact.py`: convex hull, cup/cap predicates, $S_{k,\ell}$ high-above placement, general-position checks.

Test	Result
square C_4	PASS
triangle+center max convex = 3	PASS
4-cup / 4-cap	PASS
$S_{3,3}$ size 2	PASS
$S_{4,4}$ size 6, no 4-cup/cap	PASS
high-above smoke	PASS

All 7 unit tests PASS.

4. Re-certificates: ES(3) = 3 and ES(4) = 5

4.1 ES(3) = 3

Any three non-collinear points form a triangle. Box: 4×4 grid enumeration of general-position 3-subsets — **516 checked, 0 failures**; hand lower witness of size 2 with max_convex = 2. **Status: exact.**

4.2 ES(4) = 5

Probe	Checked	Failures
6×6 grid 5-subsets (genpos)	215{,}448	**0**
Triangle + 2 interior points	4{,}941	**0**
Hand lower size 4	maxC = 3	—

Status: exact (same certificates as parent; wall-clock re-run).

5. Classical cup-cap lower witnesses

Target size 2^{n-2}; require general position and no convex n-gon (hence $\mathrm{ES}(n) \geq 2^{n-2}+1$). Construction: hand for $n = 3, 4$; recursive $S_{k,\ell}$ / S_t with genpos perturb for $n \geq 5$.

n	Target	Size	Method	Genpos	max_convex	Has C_n?	OK	Wall (s)
3	2	2	hand	Y	2	N	**PASS**	< 0.01
4	4	4	hand	Y	3	N	**PASS**	< 0.01

n	Target	Size	Method	Genpos	max_convex	Has C_n?	OK	Wall (s)
5	8	8	St+perturb	Y	4	N	**PASS**	0.00
6	16	16	St+perturb	Y	5	N	**PASS**	0.39
7	**32**	**32**	St+perturb	Y	**6**	N	**PASS**	31.04

Optional checks. Size-8 witness: max_convex $= 4 < 5$ **PASS**. Size-16: max_convex $= 5 < 6$ **PASS**. Size-32: max_convex $= 6 < 7$ **PASS** (honest general-position construction).

Headline. All five lower witnesses **PASS**. The $n = 7$ witness of size 32 certifies only ES(7) ≥ 33 — **not** ES(7) $= 33$.

6. Literature ledger refresh (ES(7))

n	ES(n)	Status	Source class
3	3	exact	trivial / box
4	5	exact	Klein / ES 1935; box
5	9	exact	classical lit; box lower only
6	17	exact	Szekeres–Peters 2006 (+SAT reverify lit); box lower only
7	**?** (conj. 33)	**OPEN**	lower ≥ 33 classical; upper from general bounds

Web check 2026-09-14 (no invented papers).

1. **arXiv:2512.24061** (Dumitru, Dec 2025) — re-checked abstract/HTML: ES(7) still first open case; SAT encoding yields **UNSAT for convex-layer-anchored subfamilies only**; pronounced runtime tails; **no full proof** that every 33-point set has a convex 7-gon. Status matches parent note.
2. **erdosproblems.com #107** — Happy Ending still listed open / falsifiable (search hit 2026-09-14).

Lit-only (not re-derived here): Suk 2017 $2^{n+o(n)}$; HMPT $2^{n+O(\sqrt{n}\log n)}$; empty-hexagon $h(6) = 30$ (Heule–Scheucher 2024; not classical ES(n)).

7. Method limits (explicit)

1. Box work is finite enumeration / constructive witnesses only.
2. ES(5) and ES(6) upper halves remain literature (not re-proved on this box).
3. Size-32 $n = 7$ witness uses the house S_t+perturb pipeline; combinatorial cup-cap honesty is asserted via genpos + no convex 7 after construction.
4. arXiv:2512.24061 is cited as **partial** external progress — house does **not** claim to have reproduced those SAT certificates.
5. No asymptotic Happy Ending claim; no Clay stamp; GATE-04 **LOCKED**.

8. Results summary (`results.json`)

Item	Value
Unit tests	7/7 PASS
ES(3) re-cert	exact (516/0)
ES(4) re-cert	exact (215448/0; tri2 4941/0)
Lowers $n = 3..7$	5/5 PASS
ES(7) box	size 32, maxC = 6, certifies ≥ 33 only
Lit ES(7)	OPEN; arXiv:2512.24061 still partial anchored UNSAT
Wall total	≈ 32.6 s
`clay_closed`	**false**
GATE-04	**LOCKED**

9. Non-claims

- Happy Ending / Erdős–Szekeres conjecture **not** solved.
- ES(7) = 33 **not** proved.
- Clay-closed **false**; finished-ToE **not** claimed; J **not** invented.
- GATE-04 collection remains **LOCKED**.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.

10. Artifacts

- Lab: `/workspace/mission-control/experiments/band1/happy_ending_es7_increment/`
- `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `es_exact.py`, `run_increment.py`, `PAPER.md`
- Shelf: `2026-09-14_Band1_Happy_Ending_ES7_increment.{md,pdf}`

Next optional (board). map-folding $2 \times n$ OR Hadamard remaining $4k$ OR cap-set $a(7)$ witness hunt OR Carreras AC length-14 verify.